

ROUND 1 CONCEPT REPORT

AutoPulse

Your car's heartbeat, read by the sensors it already has. AutoPulse listens to engine sound, vibration and ECU telemetry, extracts their signatures in time and frequency, learns what is normal for *this* car, and tells the driver what to do before a fault becomes a breakdown.



SENSOR BUS		STREAM · SOURCE · DOMAIN · STATUS	
ACOUSTIC	MEMS MIC	STFT · CNN	LISTENING
VIBRATION	BLOCK ACCEL	FFT · ENVELOPE	LISTENING
CRANK	60-2 WHEEL	ORDER · SPEED	LOCKED
TELEMETRY	OBD-II · CAN	TRENDS · BASELINE	1 HZ
SAFETY	RULE ENGINE	HARD LIMITS	ARMED

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PROBLEM STATEMENT
Honeywell · Engine data

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01

At a glance

AutoPulse turns the signals a car already produces into an early-warning health system: it extracts signatures from sound, vibration and ECU data, recognises faults with machine learning, and guides the driver in plain language.

Most breakdowns announce themselves early. A failing valve ticks, a weak cylinder slows the crankshaft for a few milliseconds, a leaking cooling system warms a little faster every day. The signals exist; nobody is reading them. AutoPulse reads them continuously, compares them with **what is normal for this particular car**, and separates **normal ageing** (shown like a phone's battery health) from **a real fault** (an alert with a cause and an action).

ENGINE STREAMS **10** + 6 vehicle Sound, vibration, knock, crank speed, ECU telemetry	MODEL FAMILIES **4** CNN · SVM · ANN · Softmax regression, as the brief asks	ENTRY COST **~₹1k** OBD-II dongle + the driver's phone. No custom sensors
FAULT CLASSES TODAY **7** + 9 vehicle Misfire, knock, valve wear, bearing, belt, imbalance…	REAL ENGINE SOUND **90.3** % SVM 321 real recordings, each held out of the fold that scored it	FALSE-ALARM DAYS **0** vs 74 Adaptive baseline vs a fixed one, over 8 simulated months

FOUR IDEAS THAT SHAPE THE PRODUCT

01 · SIGNATURES FIRST

Faults have physical fingerprints

Knock rings at the cylinder's resonance (~6–7 kHz), a misfire leaves a half-order dip in crank speed, a bearing repeats at its defect frequency. We extract these explicitly, so every alert can be explained.

02 · NORMAL IS PERSONAL

Each car learns its own baseline

A winter cold start sounds different from a summer idle, and engines age. AutoPulse learns normal per car and per context, and only flags what doesn't fit.

03 · REUSE, DON'T ADD

Sensors the car already has

About 70% of the sensing already exists behind the OBD-II port. The rest comes from the driver's phone and a cheap clip-on kit, not F1-style custom hardware.

04 · TELL PEOPLE WHAT TO DO

Guidance, not just warnings

Hard safety limits never adapt. When one trips, the driver gets the action: turn the AC off, how many minutes of battery remain, where the nearest tyre shop is.

02

The problem, as we read it

The brief asks for three things in order: extract signatures from raw multi-sensor engine data, recognise patterns with machine learning, and present both on a dashboard.

THE BRIEF SAYS	WHAT IT MEANS	HOW AUTOPULSE ANSWERS IT
Isolate frequency, time-domain or electrical patterns (signatures) from raw sensor data	The DSP core: turn raw waveforms into features that a fault changes	28 signature features per window: time statistics, FFT band energies, engine-order amplitudes, Hilbert envelope, STFT spectrogram, crank-speed irregularity (section 07)
Spot normal operational baselines	Anomaly detection trained on healthy behaviour	Adaptive per-car, per-context baseline that absorbs ageing and seasons but not faults (section 09)
Predict component wear	Slow degradation trends and remaining useful life	Health % per component with trend extrapolation, e.g. battery replacement in ~N months (section 09)
Identify real-time faults	Classify the fault now	CNN on spectrograms; SVM, ANN and Softmax on signature features; ensemble vote (section 08)
Microphones, accelerometers, ECU / OBD-II / CAN telemetry	Three data families, fused	Ten engine streams plus six vehicle streams, mapped to where they sit on the car (section 04)
Simulink / MATLAB or synthetic / open data	No hardware build is expected	Hybrid: our physics generator + public datasets, plus a MATLAB crankshaft simulation as an independent check (section 11)
Dashboard with signatures and recognised patterns	Show the evidence, not just the verdict	Startup health check, live telemetry, signature lab, baseline and wear views (section 12)

WHERE WE CORRECTED OUR OWN FIRST READING

SCOPE

Engine first, whole car second

The brief is about combustion-engine data. Brakes, tyres and battery are extensions; engine signatures come first.

INSPIRATION

Copy Tesla's UX, not its sensors

An EV has no knock, misfire or valve sound. We borrow the pre-drive check and plain-language alerts; the sensing is ours.

HARDWARE

Affordability is an argument, not a build

Data comes from simulation and public sets. The low-cost sensor tiers in section 05 show the design would survive contact with a real car.

WHAT WE LEARNED FROM F1 AND OEMS

An F1 car carries roughly 150–300 sensors and streams about a million data points a second to the pit. We take its habits, not its hardware: plausibility checks between sensors, component-life budgets, and overlaying today's signal on a reference. From road cars we take GM OnStar's proactive alerts (battery, starter, fuel pump), BMW's condition-based service, and the OBD-II rule that already checks fuel-vapour leaks by pressure decay instead of a gas sensor.

03

How AutoPulse works

Four stages: sense, extract signatures, recognise patterns, guide. Raw values also bypass the learning stages and go straight to the safety rules, so a hard limit never waits for a model.

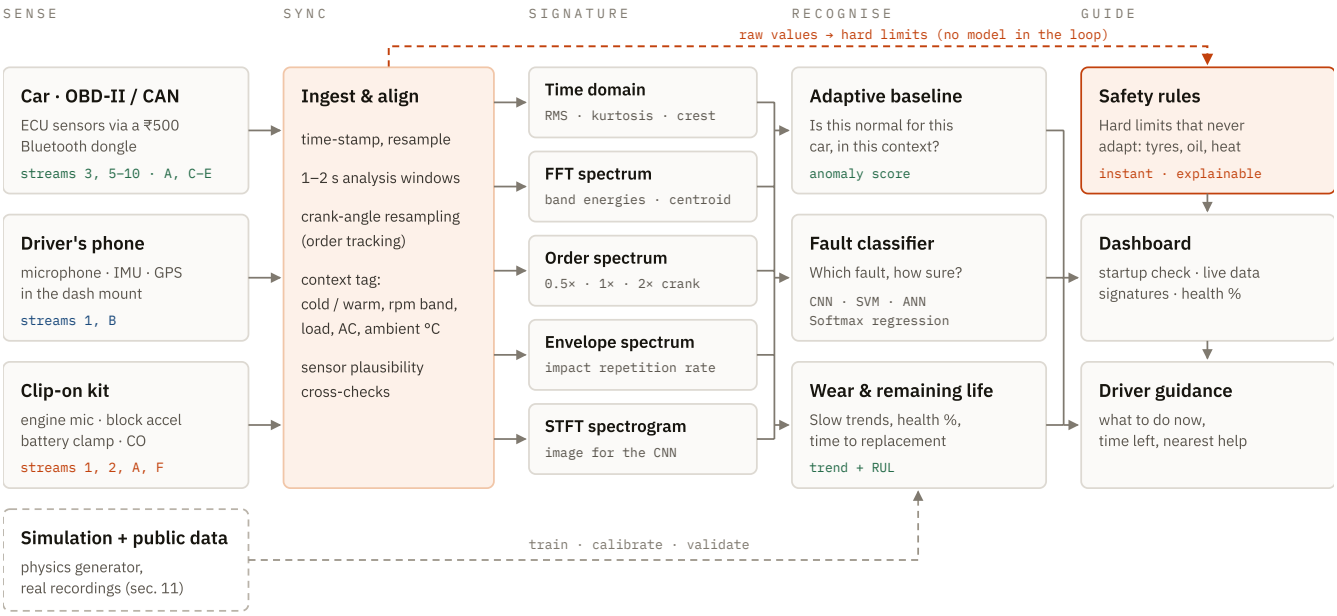


FIG 01 System architecture. Solid arrows carry live data; the orange dashed path sends raw values straight to the safety rules; the grey dashed path is offline training and calibration.

ONE WINDOW, END TO END

1. **Capture.** One second of block vibration at 24 kHz, the crank-speed pulses for that second, and the latest OBD values (RPM, load, coolant, trims).
2. **Tag the context.** 1,500 rpm, warm engine, 31 °C ambient, AC on. Context decides which baseline the window is compared against.
3. **Extract signatures.** 28 numbers plus a 64 × 95 spectrogram image.
4. **Recognise.** The baseline scores how unusual the window is *for this car*; the classifiers name the most likely fault; trends update the wear meters.
5. **Guide.** If nothing is wrong, the health score stays green. If something is, the driver gets one card: cause, confidence, and the next action.

04

Sensors: what, where and why

Ten engine streams carry the signatures the brief asks for; six vehicle streams extend AutoPulse to the rest of the car. Colour shows how each one is obtained.

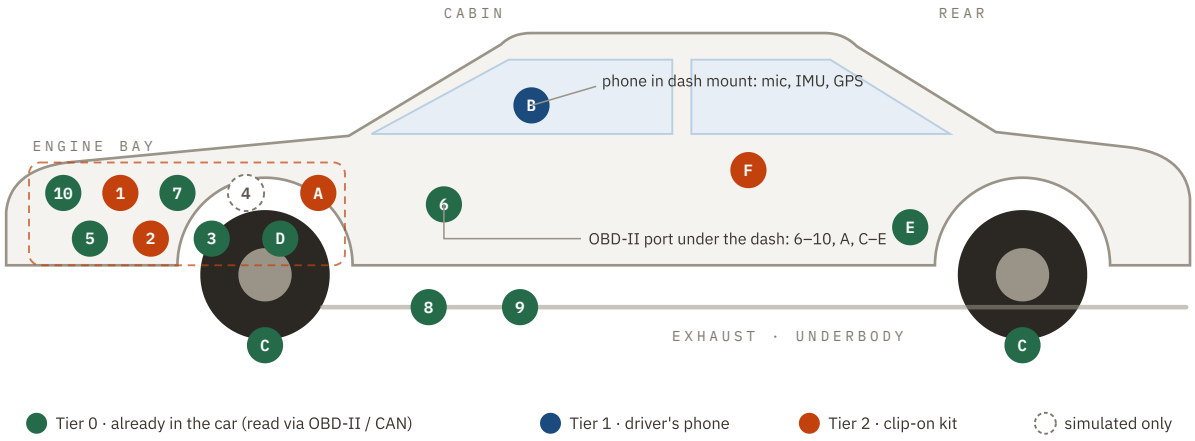


FIG 02 Where each stream is measured. Numbers 1–10 are the engine streams; letters A–F the vehicle streams in the second table.

TEN ENGINE STREAMS

STREAM	SENSOR · WHERE	WHAT IT REVEALS	RATE	SOURCE
1 **Engine sound**	MEMS microphone in the engine bay, or the phone in the cabin	Knock, tappet tick, belt squeal, exhaust leak, bearing whine	16–48 kHz	T1 T2
2 **Block vibration**	Accelerometer, magnet-mounted on the engine block	Misfire, mounts, imbalance, bearings, valve train	10–25 kHz	T2
3 **Knock sensor**	Piezo sensor already bolted to the block; read as ECU timing retard	Detonation, wrong fuel grade, carbon build-up	5–20 kHz band	T0
4 **Cylinder pressure**	Not fitted to road cars. Simulated with a Wiebe burn model as ground truth	Combustion quality; explains every other signal; labels for training	per crank °	SIM
5 **Crank speed**	60-2 tooth wheel with a Hall / VR sensor at the crank pulley	Misfire per cylinder, imbalance, timing-chain stretch (vs cam)	per tooth	T0
6 **Speed, throttle, load**	OBD-II PIDs 0C, 11, 04	Context for every signature; driver demand	1–10 Hz	T0
7 **Air flow, manifold pressure**	MAF and MAP sensors · PIDs 10, 0B	Vacuum leaks, clogged air filter, valve sealing	1–10 Hz	T0
8 **Mixture: λ and fuel trims**	Upstream / downstream O2 sensors · PIDs 06–09, 24–2B	Vacuum leaks, injectors, fuel pump, catalyst efficiency	1–10 Hz	T0
9 **Exhaust / catalyst temp**	Catalyst temperature · PIDs 3C–3F	Misfire damage risk, catalyst health, ignition retard	1 Hz	T0
10 **Coolant temperature**	ECT sensor · PID 05	Thermostat, coolant leak, overheating	1 Hz	T0

Derived (virtual) streams: **torque and shock torque** from crank speed and the pressure model, and an **emissions estimate** (NOx, CO, HC) from λ, temperatures and ignition advance.

SIX VEHICLE STREAMS

STREAM	SENSOR · WHERE	WHAT IT REVEALS	SOURCE
A **Battery voltage, cranking current**	System voltage (PID 42); clip-on battery monitor or Hall clamp	Battery health, starter wear, alternator diodes, relative compression per cylinder	T0 T2
B **Phone IMU and GPS**	Driver's phone in the dash mount	Braking effectiveness, damper wear, location for nearest help	T1
C **Wheel speed × 4**	ABS sensors on CAN, or valve-cap TPMS	Puncture, slow leak, tyre wear, wheel bearing	T0 T2
D **Oil pressure**	Existing oil-pressure sensor (manufacturer PID)	Pump or bearing wear, oil loss	T0
E **Fuel level, fuel rate, EVAP pressure**	PIDs 2F, 5E, 32	Liquid fuel leak, vapour leak, range	T0
F **Cabin CO**	Small CO sensor in the cabin	Exhaust gas entering the cabin (safety)	T2

05

Reuse before adding

A dedicated sensor for every fault would make an F1 car: precise, expensive and fragile. AutoPulse reads what the car already measures, adds a few commodity parts, and infers the rest from physics.

TIER 0 · ~₹1K
The car itself
OBD-II port + a Bluetooth dongle. ECU sensors, fault codes and the ECU's own self-test results.

TIER 1 · ₹0
The driver's phone
Microphone, accelerometer, gyro, GPS, camera. Already in every pocket.

TIER 2 · ~₹4-10K
Clip-on kit
Engine-bay MEMS mic, magnetic accelerometer, battery monitor, valve-cap TPMS, CO sensor. All off the shelf.

TIER 3 · OPTIONAL
Fleet / premium
Oil-quality sensor, decoded CAN for ABS and brake pressure, IR temperature.

VIRTUAL SENSING: INFER IT INSTEAD OF ADDING A SENSOR

INSTEAD OF…	AUTOPULSE USES…	BECAUSE…
A fuel-vapour gas sensor	EVAP pressure decay (stream E)	Seal the system, pressurise it, time the decay. A leak drops pressure faster. OBD-II already mandates this test
Tyre pressure sensors	ABS wheel speeds (C)	An under-inflated tyre has a smaller rolling radius, so it spins faster than the other three
A compression gauge	Cranking current ripple (A)	Each compression stroke loads the starter; a weak cylinder makes a smaller current hump

INSTEAD OF…	AUTOPULSE USES…	BECAUSE…
In-cylinder pressure sensors	Crank angular speed (5)	A cylinder that doesn't fire stops pushing, so the crank slows for that firing event
Opening the timing cover	Cam vs crank phase (5)	A stretched chain makes the cam lag the crank by a growing angle
A smoke test for leaks	Fuel trims vs load (7, 8)	Unmetered air at idle makes the ECU add fuel; the trim normalises under load
A thermostat test	Coolant warm-up curve (10)	A thermostat stuck open stretches the warm-up time constant
A fuel-leak detector	Fuel level vs fuel burned (E)	If the tank empties faster than the engine burns fuel, the difference is leaking
Duplicate sensors	Analytical redundancy (6, 7)	Predict MAF from MAP, RPM and throttle. If the measured MAF disagrees, the sensor itself is suspect

WHAT IF A SENSOR FAILS?

Every reading is cross-checked against a physical model built from other sensors. A disagreement is reported as a **sensor fault**, not a car fault. The system checks its own instruments, as F1 telemetry does.

06

What can go wrong

Fourteen common failures, the physics that betrays each one, and the streams that see it. Every fault is covered by at least one Tier 0 or Tier 1 stream, so the kit is an upgrade, not a requirement.

FAULT	PHYSICAL SIGNATURE	1	2	3	4	5	7	8	9	10	A	C	D	E
Misfire	Missing combustion impulse; crank slows each cycle	○	●		●	●		○	○					
Knock / detonation	Pressure ring at cylinder resonance, 5–8 kHz	●	●	●	●									
Valve train wear	Impact once per cam revolution, 2–10 kHz ring	●	●											
Timing-chain stretch	Cam lags crank by a growing angle					●								
Accessory bearing	Impacts at the defect frequency (BPFO)	●	●											
Belt slip / squeal	Narrow 2–5 kHz tone on cold start	●									○			
Imbalance / mounts	Rising 1× and 2× order vibration		●			○								
Vacuum leak	Lean trims at idle that recover under load						●	●						
Thermostat stuck open	Warm-up time constant stretches									●				
Coolant leak	Temperature rises faster than load explains								○	●				
Low oil pressure	Pressure-vs-RPM curve drifts down												●	
Battery / alternator	Deeper cranking dip; ripple on the DC bus										●			
Tyre puncture / leak	One wheel spins faster than the others											●		
Fuel leak	Tank empties faster than fuel is burned													●

● primary evidence ○ supporting evidence. Stream 6 (speed, throttle, load) is context for every row.

07

Signature extraction

Each fault changes a different view of the signal. These plots come from our own pipeline, running on an engine the models never trained on.

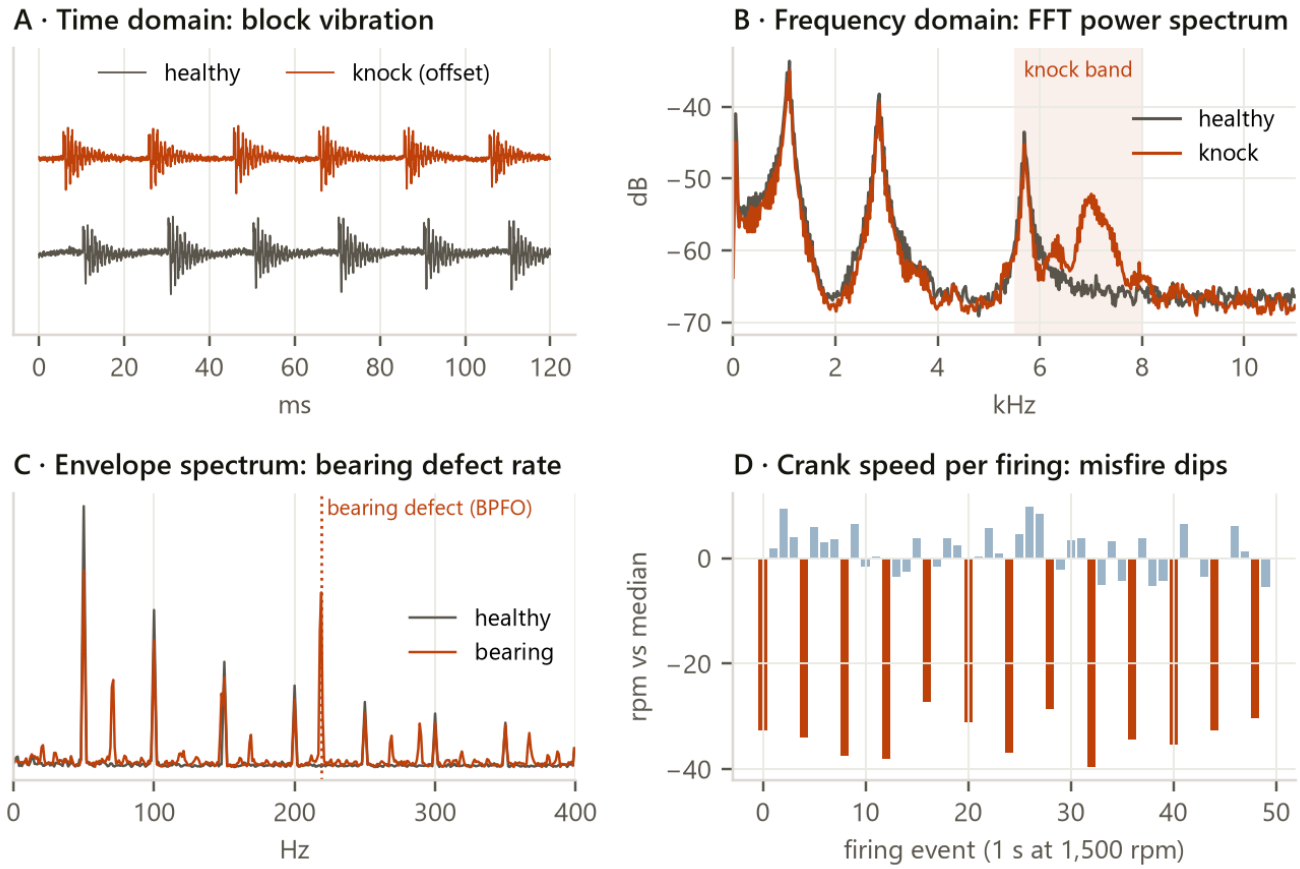


FIG 03 Four views of one second at 1,500 rpm. **A** knock adds a sharp ring after each firing. **B** that ring shows as extra energy in the 5.5–8 kHz knock band. **C** a bearing defect repeats at its outer-race frequency (BPFO), visible only after envelope demodulation. **D** a misfiring cylinder slows the crank on every fourth firing.

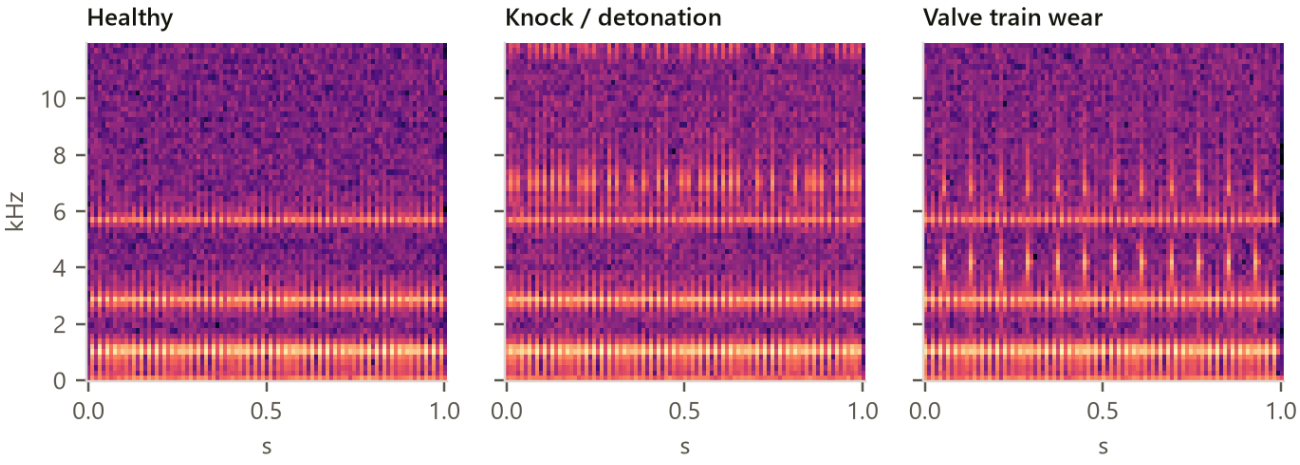


FIG 04 STFT spectrograms, the CNN's input. Knock appears as high-frequency bursts locked to firing; valve wear as one sharp vertical line per engine cycle.

DOMAIN	FEATURES	BEST AT
Time	RMS, peak, crest factor, kurtosis, skewness, zero-crossing rate	Impacts (high kurtosis), overall level
Frequency (FFT)	10 band energies, spectral centroid, flatness, knock-band kurtosis	Knock, belt squeal, resonances
Order (crank-locked)	Amplitudes at 0.5×, 1×, 1.5×, 2×, 3×, 4× engine speed	Misfire (0.5×), imbalance (1×), firing (2× on a 4-cylinder)
Envelope	Dominant repetition rate and its prominence	Bearings, valve tick
Time-frequency (STFT)	64 × 95 log-power spectrogram	CNN input: patterns that change within a cycle
Crank speed	Irregularity and deepest dip per window	Misfire, cylinder imbalance

08

Pattern recognition

All four model families the brief names are trained and scored twice: on real recordings, which is the score that counts, and on simulated engines held out of training.

ON REAL RECORDINGS

On **321 real engine recordings** (healthy, misfire, tappet, belt), the SVM classifies **90.3%** of 1-second windows and **91.0%** of whole recordings correctly, using the same 28 signature features as on the simulator. On real bearing vibration (CWRU) the SVM reaches 100.0% on a motor load it never trained on; CWRU is a well-known easy benchmark, so we treat it as a sanity check.

DATASET	MODEL	ACCURACY (± FOLD STD)	MACRO F1	PER RECORDING
Real engine sound	SVM (RBF)	90.3% ± 2.4	0.915	91.0%
	Softmax regression	88.9% ± 2.2	0.901	89.7%
	ANN (MLP + softmax)	90.2% ± 2.4	0.913	90.0%
	CNN (spectrogram)	83.0% ± 7.6	0.845	82.9%
Real bearing vibration (CWRU)	SVM (RBF)	100.0% ± 0.0	1.000	
	Softmax regression	100.0% ± 0.0	1.000	
	ANN (MLP + softmax)	98.0% ± 0.8	0.980	
	CNN (spectrogram)	100.0% ± 0.0	1.000	
Engine telemetry table	SVM (RBF)	66.1% ± 0.6	0.578	
	Softmax regression	66.2% ± 1.0	0.579	
	ANN (MLP + softmax)	66.6% ± 0.5	0.599	
	Gradient boosting (reference)	66.3% ± 0.8	0.604	

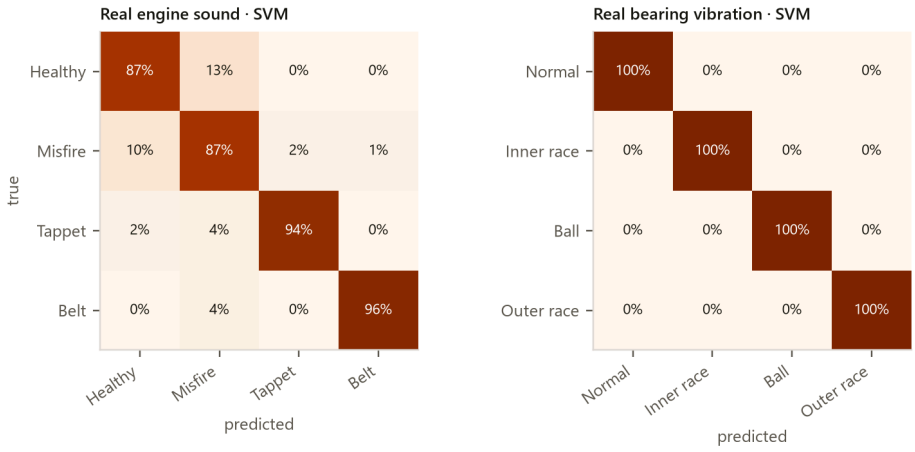


FIG 05 Best model on each real set, row-normalised. Every recording was held out of the fold that scored it.

WHAT THE REAL SCORES DO AND DON'T SHOW

- The audio set has no car or session labels, so one car may appear on both sides of a split. Scores can be optimistic until we test on a car the models have never heard.
- Models trained only on our simulator reach macro F1 0.39 at best on this real audio. The simulator must be calibrated to real sound before it can replace real data, which is round 2's first task.
- The public engine-condition table is barely learnable: the best model scores 66.3% against 63.1% for always guessing "faulty", and a healthy-only anomaly model scores AUC 0.52. Its labels don't track its sensors, so we don't use it as a benchmark.

ON THE SIMULATOR (HELD-OUT ENGINES)

MODEL	INPUT	ACCURACY	MACRO F1
SVM (RBF)	28 signature features	98.3%	0.983
ANN (MLP + softmax)	28 signature features	97.9%	0.979
Softmax regression	28 signature features	97.4%	0.974
CNN (spectrogram)	64 × 95 STFT image	74.0%	0.748

1,680 training windows from 8 simulated engines; 420 test windows from 2 different engines. The simulator is cleaner than a real engine bay, so these are upper bounds.

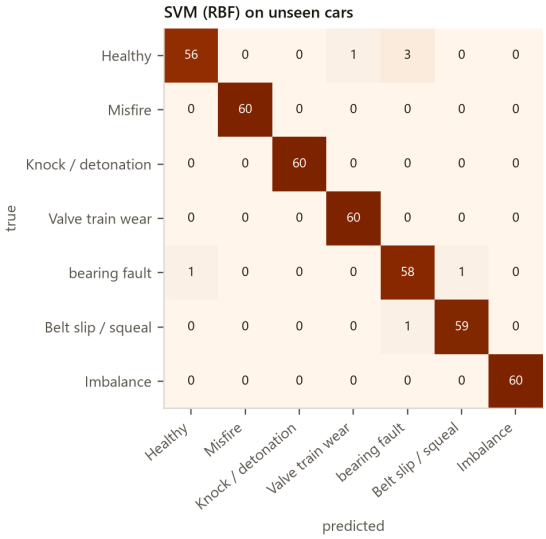


FIG 06 Best model on the two unseen simulated engines.

WHY FOUR MODELS, NOT ONE

SVM
Strong with few, well-designed features. Our current best, and easy to deploy on a phone.

SOFTMAX
Multinomial logistic regression. The transparent baseline: its weights show which signatures matter.

ANN
A small MLP with a softmax output. Learns feature interactions the linear model misses.

CNN
Learns from the spectrogram image directly. Weakest on real audio today; it needs more real data to catch up.

The dashboard shows all four votes side by side and an ensemble verdict. A fault the models disagree on is reported with lower confidence instead of being hidden.

09

Learning each car's normal

A fixed "healthy" baseline fails on real cars: cold starts get noisier in winter and engines age. AutoPulse learns normal per car and per context, and still catches a real fault within days.

LAYER 1
Population prior
What any healthy engine of this type does. Works on day one, and catches a fault that was already present when AutoPulse was installed.

LAYER 2
This car
Learned over the first ~10 drives, then updated slowly. The car's own fingerprint.

LAYER 3
This context
Cold start vs warm idle, RPM band, load, AC, ambient temperature. A cold start is compared with past cold starts.

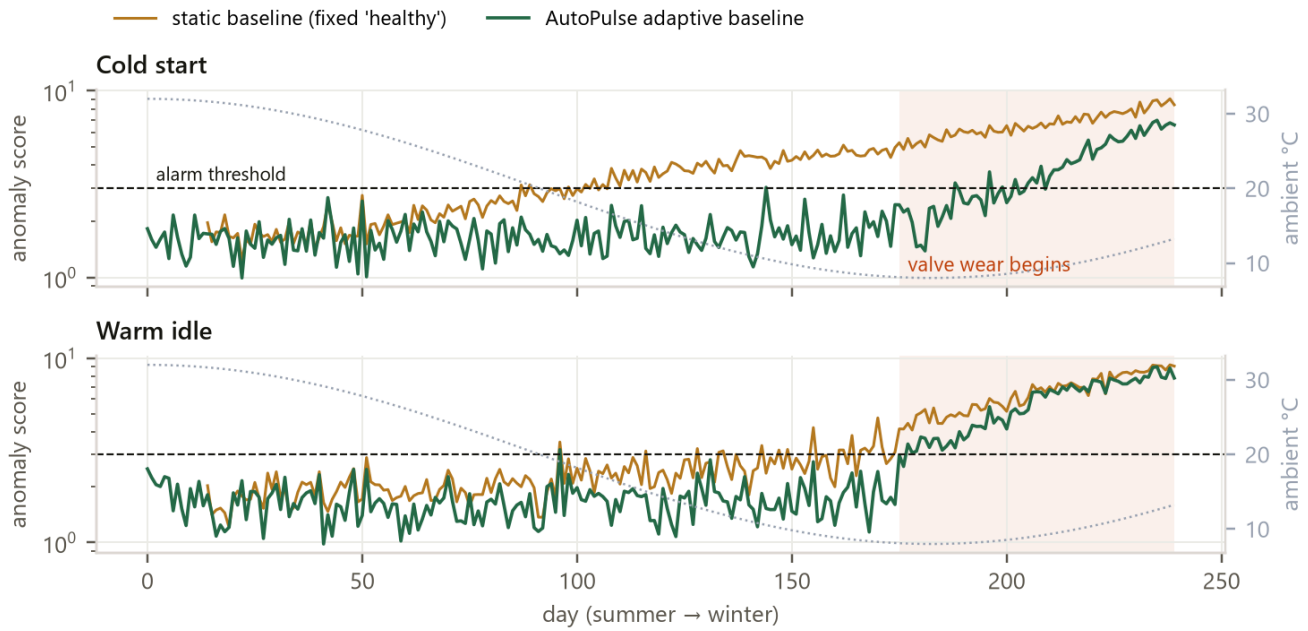


FIG 07 One simulated car over 240 days, from summer to winter, with normal ageing; valve-train wear starts on day 175. The fixed baseline crosses the alarm line through the winter, before any fault exists; the adaptive one stays below it until the fault, then rises in both contexts.

FALSE ALARMS · FIXED	FALSE ALARMS · ADAPTIVE	DETECTION · ADAPTIVE
74	0	3 days
Days with alarms before any fault existed	Seasons and ageing absorbed	From the first sign of valve wear

ON REAL DRIVING (CAROBD, TOYOTA ETIOS)

Taught by 54 real trips, with settings and thresholds fixed on 15 more, the adaptive baseline raised a false alarm on **14.3%** of 35 held-out healthy trips, against 31.4% for a fixed baseline. On the same real trips with faults injected from minute 3, it caught 91.4% of vacuum leaks and 80.0% of weak-charging faults. Small sensor drifts mostly hide inside normal driving variation for both methods; predicting each signal from the others (MAP from RPM and load, for example) is the next step.

HELD-OUT REAL TRIPS	ADAPTIVE · ALARMED	DELAY (MIN)	STATIC · ALARMED	DELAY (MIN)
Healthy trips with a false alarm	14.3%		31.4%	
vacuum leak	91.4%	1.5	91.4%	2
charging weak	80.0%	2	68.6%	2.5
map sensor drift	14.3%	0	11.4%	10.75
timing retard	11.4%	0	2.9%	12

WEAR VS ANOMALY

- **Wear** is slow, smooth and one-directional. It becomes a health % bar: "Battery 82%, replace in ~7 months".
- **An anomaly** is a step change, or drift faster than normal ageing. It becomes an alert card.

SAFEGUARDS AGAINST LEARNING A FAULT

- Learn only from windows that already look healthy
- Learn slowly (weeks, not minutes), anchored to the population prior
- Freeze every context while an anomaly is open; re-learn only after "serviced" is tapped

10

Hard limits and driver guidance

Some conditions are dangerous whatever the car's normal is. A rule engine above the models fires instantly and tells the driver what to do, using the phone's GPS for the nearest help.

CONDITION	DETECTED FROM	LEVEL	WHAT THE DRIVER IS TOLD
Tyre puncture	Pressure falling fast, or a sudden wheel-speed mismatch	CRITICAL	Slow down, hazards on, pull over. Nearest tyre shop with distance and a call button
Oil pressure lost	Oil pressure below the floor at running RPM	CRITICAL	Stop the engine now. No range shown: continuing destroys the engine in minutes
Overheating / coolant leak	Coolant level and rate of rise	WARNING	AC off; heater to max hot, fan full; estimated minutes to critical; nearest mechanic
Alternator not charging	Voltage below 13 V while running	WARNING	Minutes of driving left on the battery, with and without non-essentials; don't switch off
Serpentine belt broken	Voltage falling and coolant rising together	CRITICAL	Pull over: two belt-driven systems are failing at once
Fuel leak	Fuel level falling faster than fuel burned	CRITICAL	Stop in the open, engine off, don't restart, move away
Severe misfire	Misfire rate (crank speed, ECU self-test)	WARNING	Reduce load, avoid the highway: the catalytic converter is at risk
Carbon monoxide in cabin	Cabin CO sensor	CRITICAL	Open windows, pull over, engine off
Low fuel	Fuel level ÷ recent consumption	INFO	Range in km and the nearest fuel stations

WARNING · COOLING

Engine running hot: possible coolant leak

Coolant 104 °C, rising 1.8 °C/min. Estimated time to critical: ~4–8 min.

- Turn the AC off
- Heater to max hot, fan full
- Drive gently to the nearest mechanic

NEAREST MECHANIC · 1.4 KM →

CRITICAL · TYRES

Rapid pressure loss: front-left tyre

Front-left at 18 psi and falling 15 psi/min.

- Slow down gradually, don't brake hard
- Hazard lights on
- Pull over safely; fit the spare or call for help

NEAREST TYRE SHOP · 2.1 KM →

Alert mock-ups as the dashboard renders them. Distances are illustrative; the prototype opens a live map search. Time estimates are always shown as a range, because they are extrapolations.

11

Data: hybrid by design

No public dataset records sound, vibration, crank speed and OBD from the same faulty engine. So we simulate the full set with physics, and use real datasets to calibrate the simulator and to test every model.

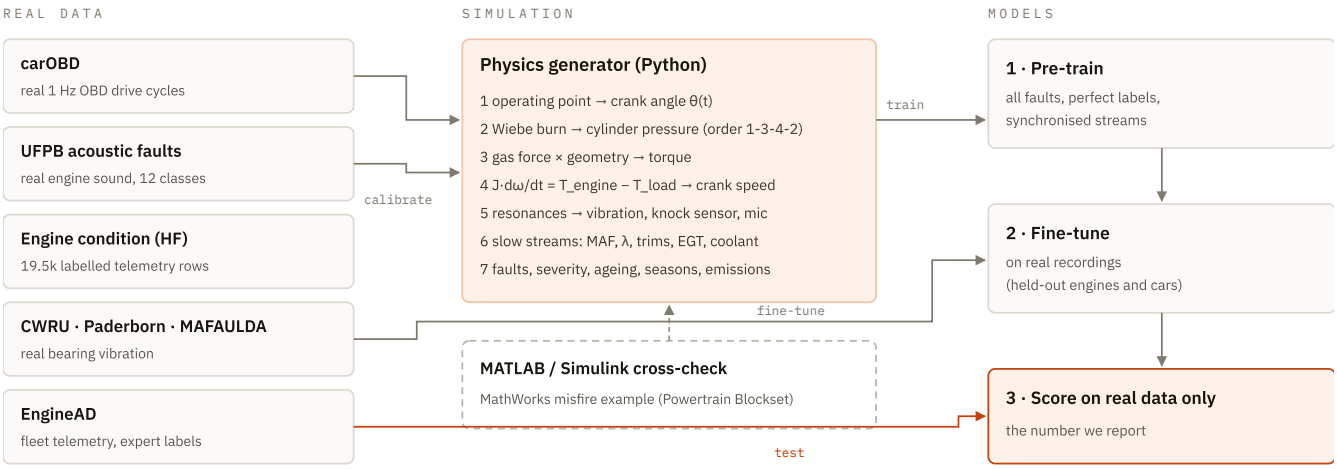


FIG 08 Hybrid data flow. Real data calibrates the generator, the generator pre-trains the models, and real data fine-tunes and finally scores them. Only real-data scores are reported as results.

DATASETS AND TOOLS, AND WHAT EACH ONE IS FOR

SOURCE	WHAT IT GIVES US	STREAMS	ACCESS
carOBD	Real 1 Hz OBD-II logs (Toyota Etios): RPM, throttle, load, coolant, trims, λ, timing, MAP, catalyst temps. Drive cycles for the generator	6–10	Free, ready today
UFPB acoustic fault dataset	Ford Fiesta 1.6 petrol, mic at 44.1 kHz, 12 classes incl. misfire and belt faults, 2,148 one-second clips	1	IEEE DataPort login
Engine condition (HF)	19,535 rows: RPM, oil and fuel pressure, coolant pressure and temp, oil temp, good/faulty label	10, D	Free; licence unstated
CWRU · Paderborn · MAFAULDA	Real bearing and rotating-machine vibration; Paderborn adds motor current, MAFAULDA a microphone	1, 2	Free (Paderborn non-commercial)
Engine main-bearing vibration	Vibration from engine main bearings across operating conditions	2	Mendeley; to verify
EngineAD	25 commercial vehicles, 6 months, 13 engine signals, expert anomaly labels	6–10	Free (Borealis)
IoT Vehicle Telemetry	3 months of RPM, throttle, coolant, fuel use	6, 10	IEEE DataPort
AudioSet "engine knocking", Activa 6G audio	Rough-labelled knock clips; healthy vs faulty scooter engine sound	1	Free / DataPort

SOURCE	WHAT IT GIVES US	STREAMS	ACCESS
MathWorks misfire example	SI engine with misfire injection, cylinder pressure vs crank angle, OBD misfire logic	4, 5	Needs Powertrain Blockset
Cantera IC engine example	Free thermodynamics to shape pressure and emission curves offline	4, emissions	Free

Dataset survey and generator design steps 1–7 come from our team's data-sources and simulation plan.

MATLAB: AN INDEPENDENT ENGINE AND AN INDEPENDENT DSP CHECK

MATLAB ENGINE SIMULATION · CRANK_SIM.M

Crankshaft physics with misfires

MATLAB solves J·dω/dt = Σ gas torque − load torque for a 4-cylinder engine (firing order 1-3-4-2) and records crank speed as a 60-2 tooth wheel would.

Result: a misfire detector trained only on our Python simulator detects 100.0% of 84 MATLAB runs, and crank physics names the misfiring cylinder in 100.0% of misfire runs, at 800–3,500 rpm.

MATLAB DSP RE-IMPLEMENTATION · SIGNATURE_CHECK.M

Same signatures, computed twice

MATLAB recomputes our signature features from first principles on real engine recordings, and Python compares them feature by feature.

Result: all 23 features agree on 60 real windows to within 6e-14 (floating-point noise); minimum correlation 1.000000.

Our college licence covers Simulink, Powertrain Blockset, AUTOSAR Blockset and Signal Processing Toolbox; installing them next unlocks MathWorks' own misfire example and the five filter-based features.

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The dashboard

A working prototype already runs end to end: generator, signatures, four models, the adaptive baseline and the safety rules, behind one dashboard.

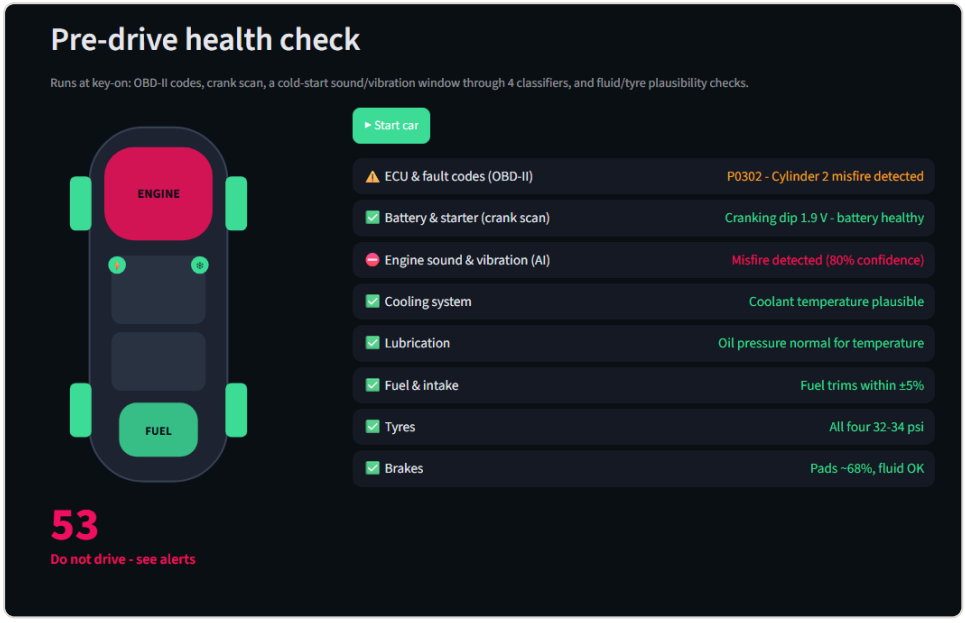


FIG 09 Pre-drive health check: key-on scan of fault codes, crank, a cold-start sound window through four classifiers, fluids and tyres, then one health score.

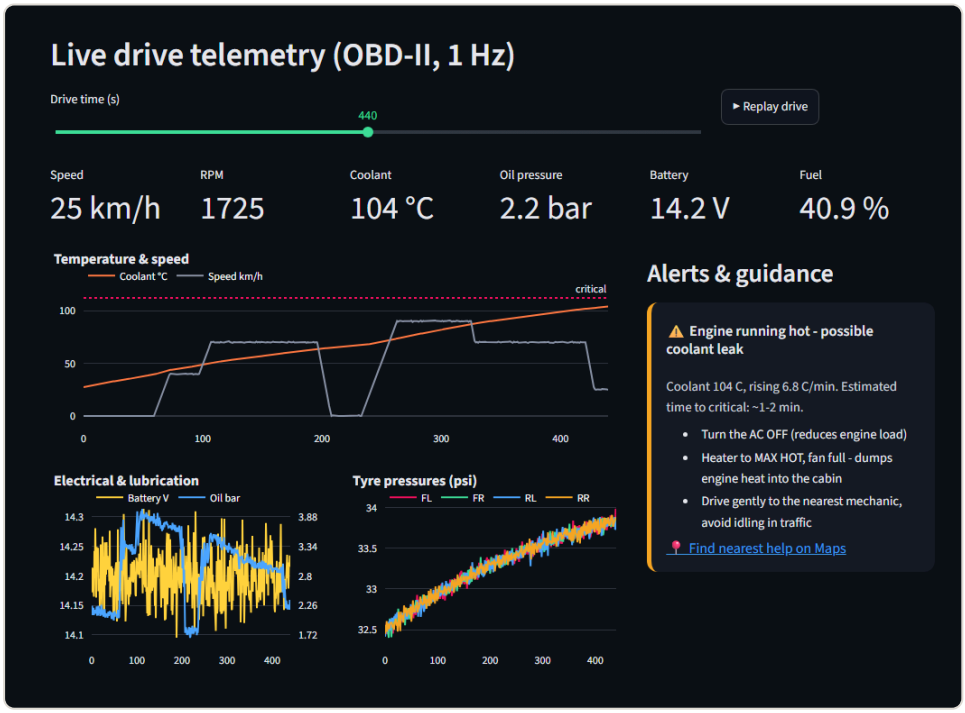


FIG 10 Live drive with a coolant leak injected: the warning appears with "AC off, heater max", the estimated time to critical, and a link to the nearest mechanic.

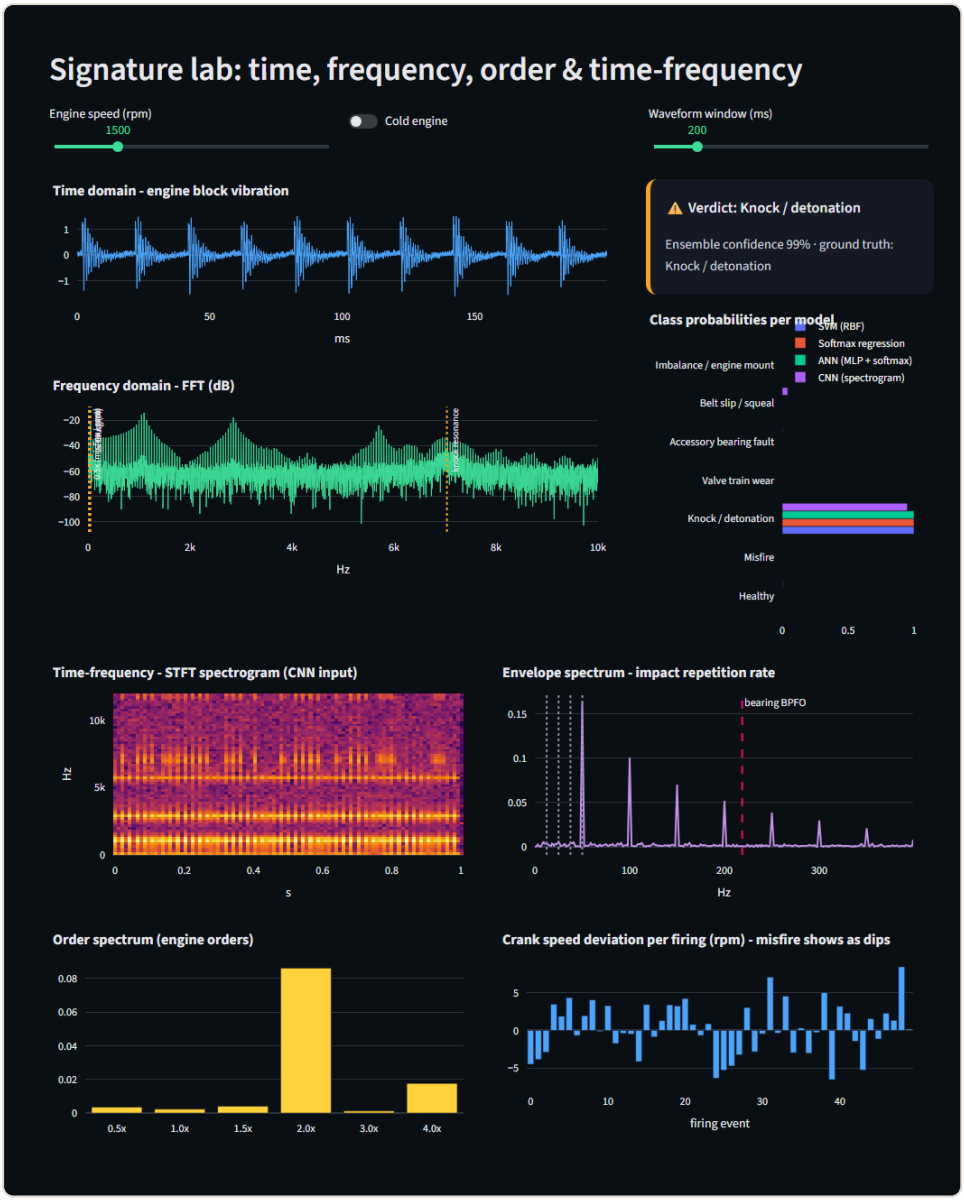


FIG 11 Signature lab: time, FFT, STFT, envelope and order views for the selected engine, with every model's vote.

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Roadmap and team

Round 1 delivers the understanding, the design and a working pipeline on synthetic data. Round 2 proves it on real data. The final shows it on a real car.

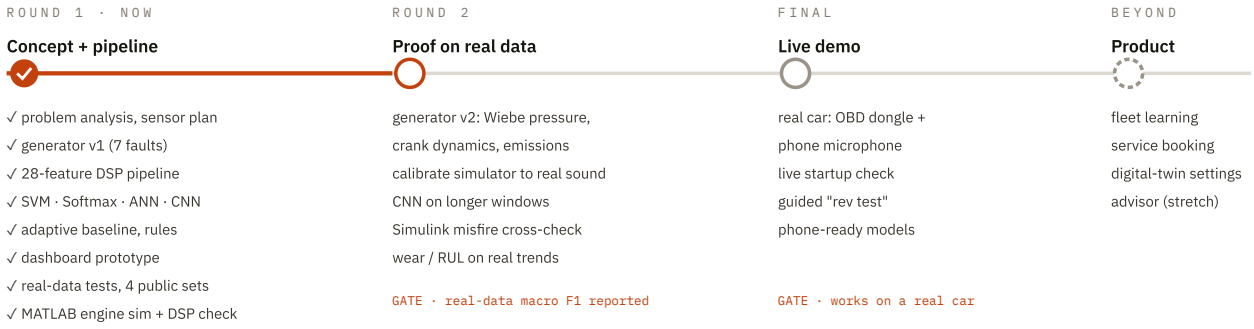


FIG 12 Roadmap. Each round ends at a gate that must be met before moving on.

WHO DOES WHAT

SIGNALS & SIMULATION

Generator v2: Wiebe pressure, crank dynamics, resonances, slow streams, emissions. Simulink cross-check.

DATA & VALIDATION

carOBD drive cycles, UFPB audio, bearing sets. Calibration and honest real-data test splits.

ML & BASELINE

Four classifiers, adaptive baseline, wear and RUL models, evaluation reports.

PRODUCT & DASHBOARD

Startup check, alerts and guidance, signature lab, the demo and this report.

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Open questions and sources

Three things would speed us up; everything else is in our hands.

1. **IEEE DataPort access.** The UFPB acoustic fault set is the best real engine-sound benchmark we found and sits behind a DataPort login.
2. **Simulink toolchain.** Our licence covers Simulink and Powertrain Blockset; we are installing them to run MathWorks' misfire example alongside our own MATLAB engine.
3. **Evaluation data.** Will the organisers provide a held-out engine dataset for the final round, or should we bring our own recordings?

SOURCES

- MathWorks, Detect misfires using on-board diagnostics
- Hojjati et al., EngineAD: a real-world vehicle engine anomaly detection dataset (2026)

- Case Western Reserve University, Bearing data center; Paderborn University, KAt bearing datacenter
- carOBD dataset (GitHub); Engine condition dataset (Hugging Face)
- Improving misfire fault diagnosis with cascading architectures via acoustic vehicle characterization, Sensors (2022)
- FenderBender, EVAP testing: working behind the scenes; Bridgestone, How TPMS works
- Jalopnik, Formula One telemetry explained; The Drive, Tesla vehicles can now diagnose problems
- BeamNG, BeamNG.tech sensors; comma.ai, opendbc